

LangGraph Fundamentals — Project Explanation

Overview

This project is the sixth entry in my Agentic AI series and introduces a third distinct multi-agent framework, LangGraph, after AutoGen's conversation-centric model and CrewAI's task and crew model. LangGraph represents agentic workflows as an explicit graph of nodes and edges, with a typed state object flowing through every step, and this notebook builds that model from a single node up to a genuine tool-using ReAct agent.

What the Notebook Does

The notebook opens with the smallest possible graph, a single typed state definition, one node function that updates a field, and a fixed edge to an end marker, establishing the core cycle every later example builds on: state flows into a node, the node returns an update, and an edge decides what runs next. It then introduces multiple nodes connected by a genuine conditional routing function, which inspects the current state and decides which of two next nodes to execute without changing the state itself, demonstrating the structural difference between processing nodes and routing functions.

The final and most substantial section builds a real ReAct-pattern agent using LangGraph's built-in message-history reducer, three genuine tools including a live DuckDuckGo web search and two arithmetic functions, and a language model bound to those tools so it can request their use during reasoning. The graph structure includes a loop-back edge from the tool-execution node back to the reasoning node, rather than terminating after one tool call, which is what allows the agent to chain multiple tool calls across multiple reasoning steps for a single user question, tested against queries that genuinely require several sequential tool calls to answer correctly.

Skills Demonstrated

- Modelling agentic workflows as explicit, typed state graphs
- Distinguishing processing nodes from conditional routing functions
- Building a genuine ReAct-pattern tool-using agent with a reasoning-execution loop
- Comparative framework literacy across AutoGen, CrewAI, and LangGraph

LangGraph's explicit, inspectable graph structure is the closest fit among the three frameworks covered in this series to the auditable, debuggable multi-agent orchestration my dissertation's agentic intelligence platform requires.